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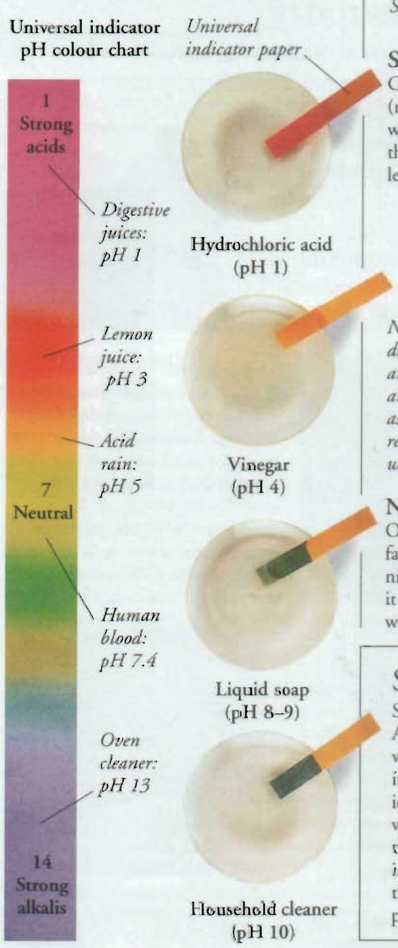
ACIDS AND ALKALIS

LEMON JUICE AND VINEGAR taste sour because they contain weak acids. An acid is a substance that dissolves in water to form positively charged particles called hydrogen ions (H^+). The opposite of an acid is an alkali, which dissolves in water to form negatively charged ions of hydrogen and oxygen, called hydroxide ions (OH^-). Alkalis are "anti-acids" because they cancel out acidity. Toothpaste, for example, contains an alkali to cancel out acidity in the mouth that would otherwise damage teeth.

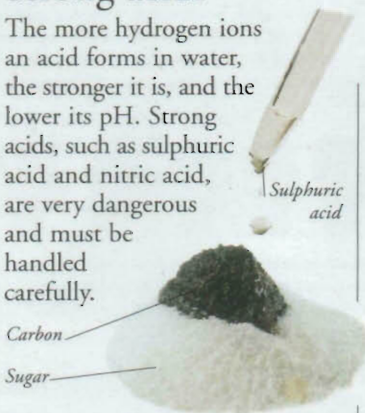
pH scale

The concentration of hydrogen ions in a solution is known as its pH. Scientists use the pH scale to measure acidity and alkalinity. On the pH scale, a solution with a pH lower than 7 is acidic, and a solution with a pH greater than 7 is alkaline. Water is neutral, with a pH of 7. A solution's pH can be tested with universal indicator solution or paper, which changes colour in acids and alkalis.



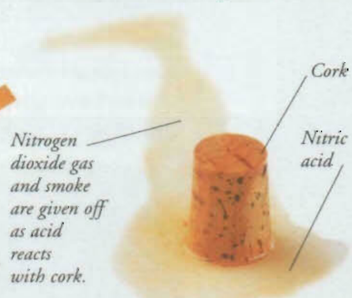
Strong acids

The more hydrogen ions an acid forms in water, the stronger it is, and the lower its pH. Strong acids, such as sulphuric acid and nitric acid, are very dangerous and must be handled carefully.



Sulphuric acid

Concentrated sulphuric acid will dehydrate (remove water from) any substance with which it comes into contact. For example, the acid dehydrates sugar, a carbohydrate, to leave a mass of smouldering black carbon.



Nitric acid

Organic matter, such as paper, cork, rubber, fabric, and skin, is rapidly decomposed by nitric acid. The acid is so corrosive because it oxidizes (supplies oxygen to) any material with which it comes into contact.

Svante Arrhenius

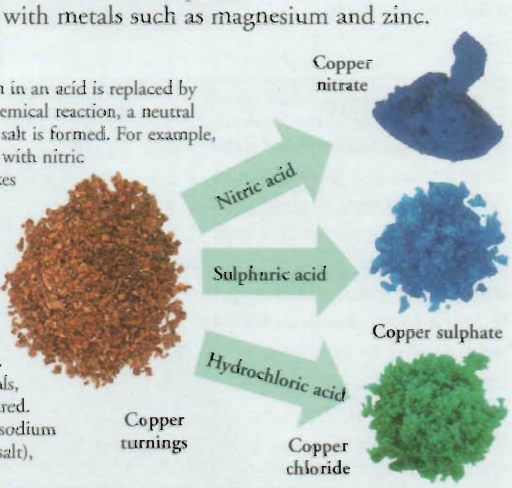
Swedish scientist Svante Arrhenius (1859–1927) won acclaim for his research into how compounds form ions in solution. This work led him to realize that it is hydrogen ions that give acids their special properties.

Acids and metals

Even the weakest acids cannot be stored in metal containers because acids are corrosive to most metals. When an acid reacts with a metal, hydrogen gas is given off and the metal dissolves in the acid to form a compound called a salt. The reaction is very violent with metals such as potassium and sodium, and quite vigorous with metals such as magnesium and zinc.

Salts

When the hydrogen in an acid is replaced by a metal during a chemical reaction, a neutral compound called a salt is formed. For example, when copper reacts with nitric acid, the copper takes the place of the hydrogen to make the salt copper nitrate. Like other metals, copper forms a variety of salts when mixed with different acids. Most salts are crystals, and many are coloured. Some salts, such as sodium chloride (common salt), occur naturally.



Acid industry

Acids are widely used in industry because they react so readily with other materials. For example, sulphuric acid is used in the production of dyes and pigments, artificial fibres, plastics, soaps, and explosives. The acid is made by sulphur and oxygen reacting together.

Sulphuric acid chemical plant

Acid rain

Burning fossil fuels to produce energy for use at home and in industry releases polluting gases into the air. The gases dissolve in water in the clouds to form nitric acid and sulphuric acid. This water falls as acid rain, which erodes stone buildings and statues, kills trees and aquatic life, and reduces the soil's fertility.